

Fast Food Marketing Campaign A/B Test

This is an analysis of the [Fast Food Marketing Campaign A/B Test](#).

Goal of the Test

The main goal of this A/B test is to identify which of the three marketing campaigns is most effective in driving sales for a newly introduced fast-food menu item. The item is launched across multiple randomly selected locations, with each location assigned exactly one promotion strategy. Sales performance is measured using weekly sales data collected over the first four weeks.

Since the dataset is aggregated by **LocationID**, **PromotionID**, and **Week**, sales data will first be aggregated at the **LocationID–PromotionID** level before conducting statistical analysis. The effectiveness of the marketing campaigns will then be evaluated through **pairwise comparisons** between promotions to determine which strategy performs best.

Because multiple statistical tests are required, a stricter confidence level of **99%** will be used to reduce the risk of Type I errors (false positives) when selecting the best-performing marketing campaign.

Target Metric

The primary target metric for this A/B test is **total sales of the new menu item**, measured in units sold. Sales data is recorded on a weekly basis for each location during the first four weeks after the product launch.

For analysis, weekly sales will be **aggregated by LocationID and PromotionID** to obtain the total sales per location for each marketing campaign. The average total sales per location will then be compared across the three promotions using pairwise statistical tests. This metric directly reflects the effectiveness of each marketing strategy in driving customer purchases of the new product.

Calculations and Data Aggregation

Before conducting statistical tests, the dataset was aggregated to the appropriate level. Although the original data is recorded weekly, the unit of analysis for this experiment is **each location**, since each location was exposed to only one promotion.

Aggregation Approach

- Weekly sales were **summed across the four-week period**
- Data was aggregated by **LocationID** and **PromotionID**
- This resulted in one total sales value per location per promotion

Summary Statistics by Promotion

After aggregation, summary statistics were computed for each promotion to understand overall performance and variability:

Promotion	Mean Total Sales	Standard Deviation	Number of Locations
Promotion 1	232.40	64.11	43
Promotion 2	189.32	57.99	47
Promotion 3	221.46	65.54	47

Table 1. Summary Statistics by Promotion
(Sales values are measured in thousands.)

Initial Observations

- **Promotion 1** shows the highest average total sales per location.
- **Promotion 2** has the lowest mean performance among the three campaigns.
- Sales variability is comparable across promotions, suggesting similar dispersion.
- Sample sizes across promotions are reasonably balanced, supporting fair comparisons.

These aggregated results form the basis for the **statistical tests**, which will be conducted next using a **99% confidence level** to account for multiple comparisons.

Hypotheses Formulation

The objective of this A/B test is to compare the effectiveness of three different marketing promotions in terms of total sales per location. Since there are **three promotions**, the analysis requires **pairwise comparisons** between each pair of campaigns.

Let μ_i represent the **mean total sales per location** for Promotion i .

Promotion 1 vs Promotion 2

- **Null Hypothesis (H_{012}):**
 $\mu_1 = \mu_2$
There is no statistically significant difference in mean total sales between Promotion 1 and Promotion 2.
 - **Alternative Hypothesis (H_{112}):**
 $\mu_1 \neq \mu_2$
There is a statistically significant difference in mean total sales between Promotion 1 and Promotion 2.
-

Promotion 1 vs Promotion 3

- **Null Hypothesis (H_{013}):**
 $\mu_1 = \mu_3$
There is no statistically significant difference in mean total sales between Promotion 1 and Promotion 3.
 - **Alternative Hypothesis (H_{113}):**
 $\mu_1 \neq \mu_3$
There is a statistically significant difference in mean total sales between Promotion 1 and Promotion 3.
-

Promotion 2 vs Promotion 3

- **Null Hypothesis (H_{023}):**
 $\mu_2 = \mu_3$
There is no statistically significant difference in mean total sales between Promotion 2 and Promotion 3.
- **Alternative Hypothesis (H_{123}):**
 $\mu_2 \neq \mu_3$
There is a statistically significant difference in mean total sales between Promotion 2 and Promotion 3.

Significance Level

Because multiple pairwise comparisons are performed, a more stringent significance level is applied to account for the increased likelihood of Type I errors. All statistical tests are therefore evaluated using a **99% confidence level ($\alpha=0.01$)**.

Consideration of Market Size

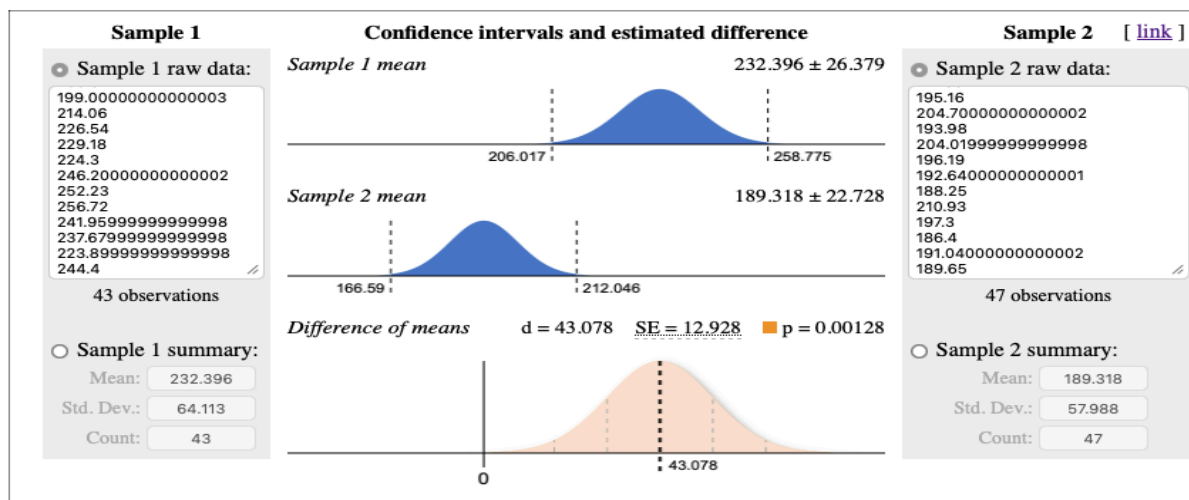
Market size was not included as a factor in the analysis because promotions were randomly assigned across locations, and the objective was to compare overall sales performance between campaigns. The total sales metric already captures variation across market sizes, making additional stratification unnecessary for the primary decision.

Statistical Testing and Results

To determine whether the observed differences in sales between marketing campaigns are statistically significant, [2 Sample T-Test](#) were conducted on the aggregated total sales per location. Each test compares the mean total sales between two promotions.

All tests are evaluated at a **99% confidence level** to account for the multiple comparisons performed.

Promotion 1 vs Promotion 2



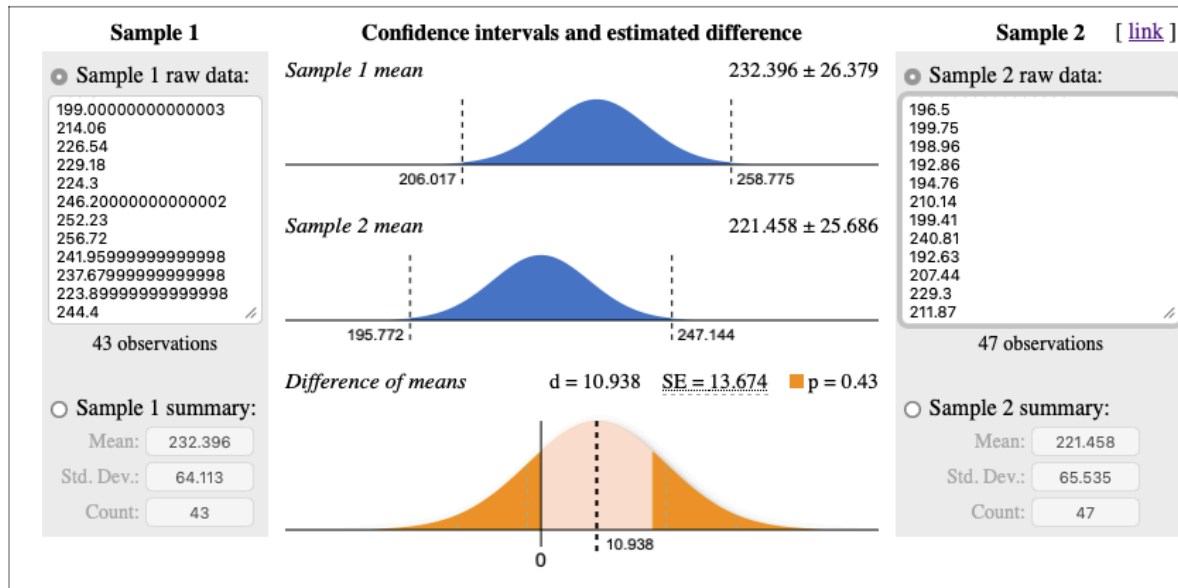
Verdict: Sample 1 mean is greater

Hypothesis: ☒ $d = 0$ ☐ $d \leq 0$ ☐ $d \geq 0$

Confidence: 99%

- **Observed result:** Promotion 1 has a higher mean total sales per location than Promotion 2.
- **Statistical outcome:** The difference in means is statistically significant at the 99% confidence level.
- **Conclusion:** The null hypothesis is rejected. Promotion 1 performs significantly better than Promotion 2.

Promotion 1 vs Promotion 3



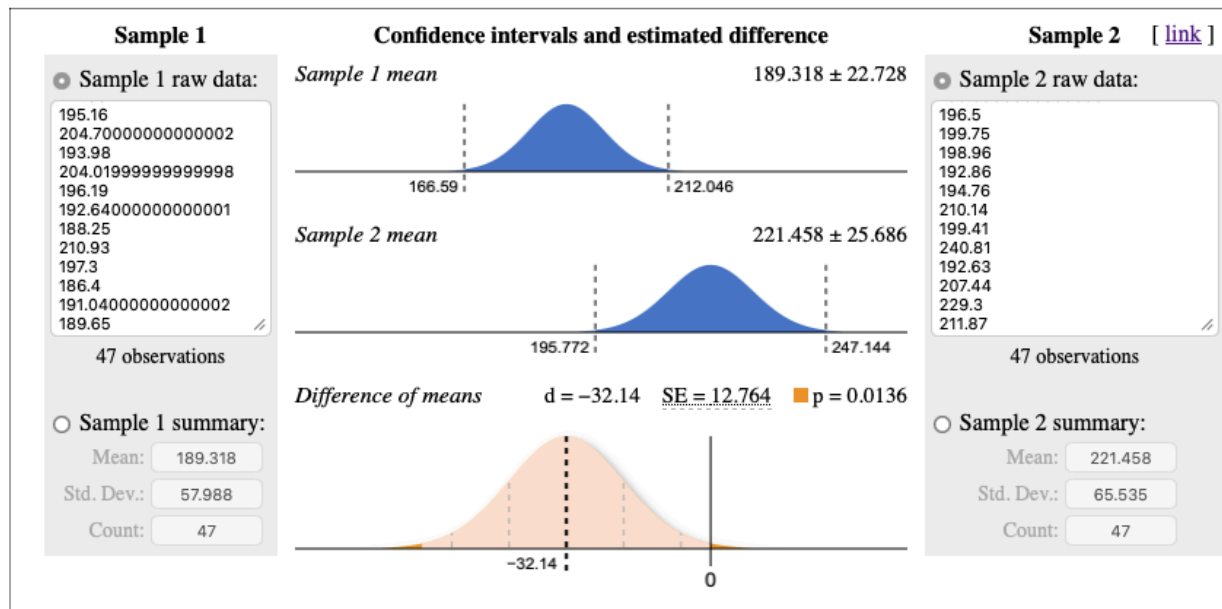
Verdict: No significant difference

Hypothesis: ☒ $d = 0$ ☐ $d \leq 0$ ☐ $d \geq 0$

Confidence: 99%

- **Observed result:** Promotion 1 shows a slightly higher average sales performance than Promotion 3.
- **Statistical outcome:** The difference is **not statistically significant** at the 99% confidence level.
- **Conclusion:** The null hypothesis cannot be rejected. There is insufficient evidence to conclude that Promotion 1 outperforms Promotion 3.

Promotion 2 vs Promotion 3



Verdict: No significant difference

Hypothesis: ☒ d = 0 ☐ d ≤ 0 ☐ d ≥ 0

Confidence: 99%

- **Observed result:** Promotion 3 has substantially higher mean sales than Promotion 2.
- **Statistical outcome:** The difference is statistically significant at the 99% confidence level.
- **Conclusion:** The null hypothesis is rejected. Promotion 3 performs significantly better than Promotion 2.

Overall Interpretation

- **Promotion 2** consistently underperforms relative to both Promotion 1 and Promotion 3.
- **Promotion 1 and Promotion 3** show comparable performance, with no statistically significant difference between them at the 99% confidence level.
- From a statistical perspective, **Promotion 2 can be eliminated** as the least effective strategy.

Final Recommendation

- Based on the results of the [2 Sample T-Test](#) conducted at a 99% confidence level, **Promotion 1 emerges as the best-performing marketing campaign**. While the difference in average sales between Promotion 1 and Promotion 3 is not statistically significant, Promotion 1 generates **significantly higher sales than Promotion 2** ($p = 0.00128$). Additionally, although Promotion 3 outperforms Promotion 2 in terms of mean sales, this difference does not reach statistical significance at the 99% confidence level.
- Overall, **Promotion 2 consistently underperforms** relative to the other two campaigns and can be ruled out as the least effective strategy. Given that Promotion 1 achieves the **highest average total sales per location** and demonstrates a statistically significant advantage over Promotion 2, it is recommended as the preferred marketing campaign for promoting the new menu item.

Appendix

Query for table 1

```
WITH location_level_sales AS (  
  SELECT  
    location_id,  
    promotion,  
    SUM(sales_in_thousands) AS total_sales  
  FROM `tc-da-1.turing_data_analytics.wa_marketing_campaign`  
  GROUP BY  
    location_id,  
    promotion  
)  
  
SELECT  
  promotion AS promotion,  
  ROUND(AVG(total_sales), 2) AS mean_total_sales,  
  ROUND(STDDEV(total_sales), 2) AS stddev_total_sales,  
  COUNT(DISTINCT location_id) AS number_of_locations  
FROM location_level_sales  
GROUP BY  
  promotion  
ORDER BY  
  promotion;
```

Query for the raw data for the [2 Sample T-Test](#).

```
SELECT  
  location_id,  
  promotion,  
  SUM(sales_in_thousands) AS total_sales  
FROM `tc-da-1.turing_data_analytics.wa_marketing_campaign`  
GROUP BY  
  location_id,  
  promotion  
ORDER BY  
  promotion,  
  location_id;
```